ComfyUI + Krea 2: what everything actually does

Plain English. No definitions that define nothing. Every part and setting gets the same three answers.

What is it?
In words, not more acronyms.

What does it do?
To your picture, or your wait.

What do I set it to?
A real starting value.

Sized for `16 GB VRAM` `32 GB RAM` with notes for smaller cards

1 The big picture READ THIS FIRST

Krea 2 turns words into pictures, and it is built for looks. Like every model of its kind, it starts with a screen of pure static and cleans it up a little at a time until it matches your prompt. The Turbo version does it in 8 passes.

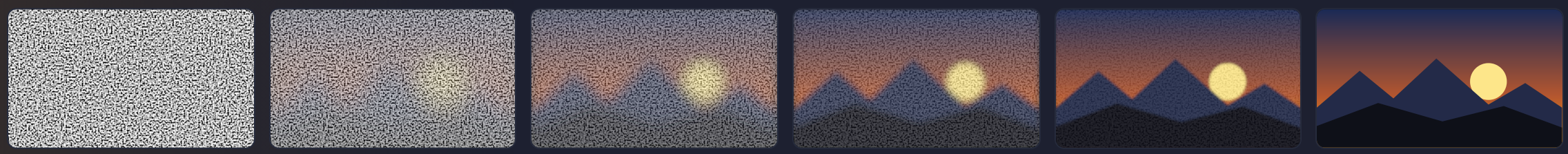
YOU
Prompt
Your words. By default the workflow rewrites them into a longer prompt first.

THE READER
Text encoder
Reads the prompt and turns it into numbers the artist can follow. Runs once per prompt.
`Load CLIP`

THE ARTIST
Diffusion model
Starts from static and cleans it up one step at a time, steered by the reader's numbers. Nearly all of your wait is here.
`Load Diffusion Model`

THE UNPACKER
VAE
The artist works on a small compressed version. The VAE turns it back into real pixels.
`Load VAE`

RESULT
Image
1024 x 1024 by default, up to 2K.



`start: static` `step 4` `step 8` `step 12` `step 16` `step 20: done`

A step is one cleanup pass. More passes, cleaner result, longer wait. (A drawing of the idea, not real output.)

Everything else on this page is one of three things: **a part** of that assembly line, **a file** that one of the parts loads, or **a knob** on the artist.

2 What it does TWO VERSIONS, THREE WORKFLOWS

Krea 2 makes pictures from text. **The official workflows do not edit pictures.** What you choose is which version of the model to load, and whether to steer its style.

Turbo: the one you generate with

A version of the model taught to finish in **8 steps**. Fast, and tuned to look good straight away. Every official ComfyUI workflow uses it. If you are not training anything, this is your file.

Raw: the one you train on

The base model before that speed training. It needs the full number of steps, about 52, so every picture takes several times longer. It exists so people can train LoRAs and fine-tunes on a clean starting point. You can skip the download.

WORKFLOW	YOU GIVE IT	WHAT IT DOES WITH IT	REACH FOR IT WHEN
`Text to image`	A prompt. Nothing else.	Invents the whole picture in 8 steps.	Always. This is the main event.
`Style LoRA`	A prompt, plus one of nine official style files switched on.	Makes the same kind of picture in that file's look: ink wash, stippling, watercolour, retro anime and so on. The template adds the style's trigger words to your prompt for you.	You want a consistent, named look across many pictures.
`Style reference`	A prompt, plus one or two pictures whose look you like.	Borrows the **style** of your pictures, not their content: the palette, the brushwork, the mood. The subject still comes from your prompt.	You have an example of the look and no LoRA for it. It needs the `int8_convrot` Turbo file and the style reference LoRA.

3 The parts WHAT EACH FILE YOU LOAD IS FOR

A workflow loads **three model files**. Each one has its own loader box on the canvas.

Diffusion model THE ARTIST

`also called: checkpoint, UNet, DiT | field: unet_name`

WHAT IT DOES

Makes the picture. This **is** the model: about 12 billion numbers.

WHY YOU CARE

It is the biggest file and it works on every step, so its size against your card decides how fast you go. Choosing this file is also how you choose Turbo or Raw.

Text encoder THE READER

`also called: CLIP | field: clip_name`

WHAT IT DOES

Reads your prompt and hands the artist numbers that mean the same thing. In the official workflow it does a second job first: it rewrites your prompt (see below).

WHY YOU CARE

It is a small language and vision model (Qwen3-VL, 4 billion parameters), so it loads quickly. Change only the seed and ComfyUI reuses its last answer.

Set the loader's type to `krea2`.

VAE THE UNPACKER

`field: vae_name`

WHAT IT DOES

Turns the artist's compressed result into real pixels.

WHY YOU CARE

Barely. It is tiny. Krea 2 borrows the one from Qwen Image: `qwen_image_vae.safetensors`. If you already have that file, you do not need another copy.

The switch that changes your results the most

Prompt enhancement ("Refine Prompt")

Before anything is drawn, the workflow asks the reader to rewrite your prompt into a much longer, more detailed one, and **that** is what gets drawn. It is **on by default**. On, a five-word prompt still gives a rich picture, but the rewriter decides everything you did not say: the lighting, the angle, the background. Off, the model draws exactly your words, so short prompts come out plain and long ones come out the way you wrote them. If a result surprises you, this switch is the first thing to check.

Add-ons and words you will keep seeing

LoRA

A small patch file that sits on top of the artist and bends it toward something: a style, a face, a subject. **Strength** is how hard it bends. 0 is off, 1.0 is what its maker intended, and well past that the picture overcooks. It has to be made for Krea 2.

Trigger words

A phrase a LoRA was trained to respond to. Without it in the prompt, the LoRA may do little. The official template looks up the phrase for the style you pick and adds it for you. In other workflows you type it yourself.

Checkpoint

Just means a saved model file. People use it loosely for the diffusion model.

.safetensors

The container, like .zip or .mp4. It tells you nothing about what is inside. The name is literal: the file can only hold numbers, so it cannot hide a program the way the older .ckpt and .pt files could.

4 File names WHICH ONE TO DOWNLOAD

Read a model's filename left to right and it tells you exactly what you are getting.

`krea2` `_turbo` `_fp8_scaled` `.safetensors`

krea2
The model family.

turbo
Which version. `turbo` is the 8-step one you generate with. `raw` is the base for training.

fp8_scaled
How its numbers are stored. This is the part that sets the file size.

.safetensors
The container. Ignore it.

The one idea behind all the letters

A model is a giant list of numbers. The letters say **how many bits each number is stored with**. Fewer bits: smaller file, more of it fits on your card, slightly less exact result. Same trade as JPEG quality or MP3 bitrate.

bf16
16 bits per number. The original, full-quality file, and the biggest one.

fp8_scaled
8 bits per number, with a scale factor kept for each layer so less is lost in the rounding. Half the size of the 16-bit original. It is the long-standing 8-bit format and runs on nearly any setup.

int8
8 bits per number, stored as whole numbers. About half the size of the 16-bit original.

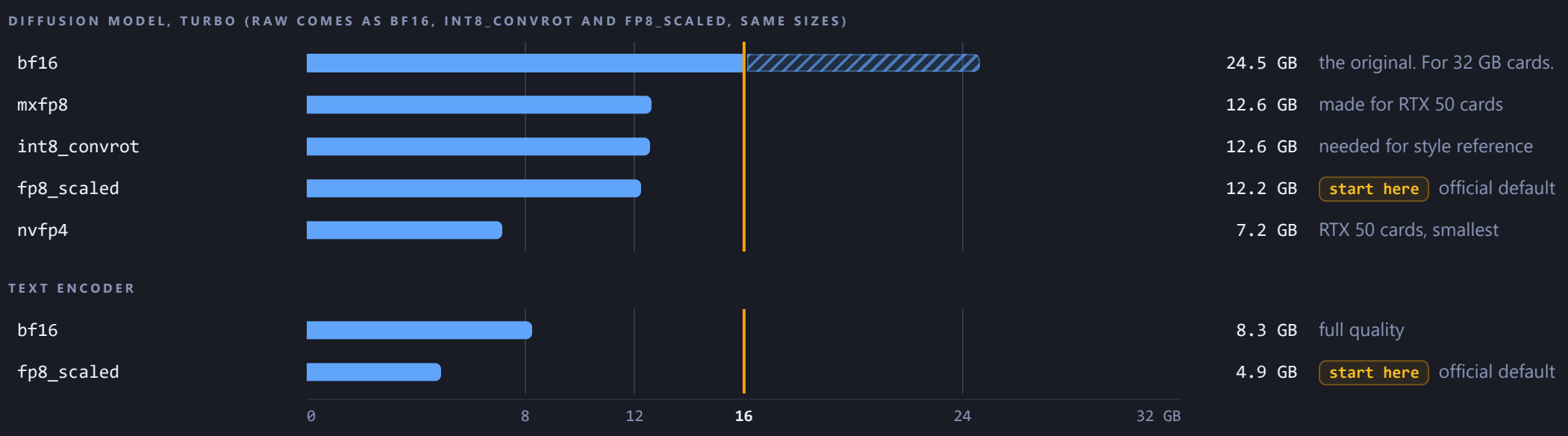
convrot
The trick that makes int8 hold up. Before rounding, the numbers get shuffled so a few extreme values cannot wreck the precision of the rest. You set nothing. Read it as "the good int8". It wants an up-to-date ComfyUI.

mxfp8
An 8-bit format that keeps a scale for every small block of numbers. Like nvfp4, it is made for RTX 50 cards with a recent PyTorch.

nvfp4
NVIDIA's 4-bit format. Less than a third the size of the original. It is made for RTX 50 cards, which can do the math on it directly.

File size against your card

fits on a 16 GB card · past 16 GB, spills into system RAM · 16 GB · faint lines: 8, 12 and 24 GB



Sizes are the official files. A model also needs working room on the card while it runs, more of it at 2K than at 1K, so a file that sits right under your line is tighter than it looks.

Download this set

PART	FILE	SIZE	GOES IN COMFYUI/MODELS/
Diffusion model	`krea2_turbo_fp8_scaled.safetensors`	12.2 GB	`diffusion_models`
Text encoder	`qwen3vl_4b_fp8_scaled.safetensors`	4.9 GB	`text_encoders`
VAE	`qwen_image_vae.safetensors`	0.2 GB	`vae`
Style LoRA optional, nine to choose from	`krea2_darkbrush.safetensors`	0.4 GB	`loras`
For style reference optional, both files	`krea2_turbo_int8_convrot.safetensors` `krea2_style_reference.safetensors`	12.6 GB 0.4 GB	`diffusion_models` `loras`

The first three are the exact set ComfyUI's built-in Krea 2 text to image template asks for. All of it is at huggingface.co/Comfy-Org/Krea-2. The model card lists the Krea 2 Community License, so read it before any commercial use.

Smaller is not automatically better. A smaller file helps for one reason: more of it fits on the card. Every step down in bits costs a little exactness, so go only as small as you have to. On a 16 GB card the default already fits. The 4-bit `nvfp4` file is for RTX 50 owners who need the room.

5 Settings THE KNOBS ON THE ARTIST

Turbo has very few knobs worth touching, which is the point of it. Change **one at a time, with the seed fixed**.

KNOB	WHAT IT ACTUALLY DOES	TURN IT UP AND	START AT
Resolution megapixels, or width x height	How many pixels are in the picture. The template sets it as megapixels plus a shape (square, wide, tall), and works out width and height for you.	The picture gets sharper and holds more small detail. Time and memory climb at least as fast as the pixel count: 2K has four times the pixels of 1K. The official workflow covers 1K to 2K.	`1 MP = 1024 x 1024` Go up once it runs and you want the detail.

Seed	The exact pattern of static the artist starts from. It is a number that picks a pattern, nothing more.	There is no up or down, only different. Same seed with everything else the same gives you the same picture again. A new seed is a new take on the same prompt.	`fixed`
			While you test a setting.
			`random`
			When you just want more takes.

Steps	How many cleanup passes the artist gets.	On most models, cleaner results for a longer wait. Turbo is different: it was taught to finish in exactly 8, so this is not a quality dial.	`8` Leave it. (Raw is the version that wants about 52.)
CFG	On other models: how hard to push toward the prompt, and the thing that makes negative prompts work. It runs the artist twice per step.	Does not apply. Turbo has that push trained into it, so the official workflows leave CFG at 1, which means off.	`1` Leave it. There is no negative prompt to write.
Sampler	The method the artist uses on each pass to decide what to clean up next.	Not a more or less knob. A different sampler reaches a slightly different result in the same number of steps.	`euler` The official pick.
Scheduler	The pacing. It decides how much static comes off at each step: big chunks early, when the layout gets decided, and fine touches late.	Not a more or less knob either. Different pacing, same number of steps.	`simple` Leave it.
Refine Prompt prompt enhancement	Lets the reader rewrite your prompt into a longer one before the picture is made.	Not a dial, a switch. On: richer pictures from short prompts, less control. Off: exactly what you typed.	`on` to explore `off` when you need it to follow your words
LoRA strength	How hard a LoRA bends the artist.	Stronger style, then overcooked: the style swallows the subject.	`1.0` The recommended value for the official style LoRAs. Drop toward 0.8 for a lighter touch.

The nine official style LoRAs and their trigger words

FILE	TRIGGER WORDS	FILE	TRIGGER WORDS
`krea2_darkbrush`	monochrome ink wash style	`krea2_retroanime`	purple retro anime style
`krea2_dotmatrix`	monochrome stippling style	`krea2_softwatercolor`	art deco watercolor style
`krea2_kidsdrawing`	naive expressive sketch style	`krea2_sunsetblur`	ethereal motion blur style
`krea2_neondrip`	textured abstract style	`krea2_vintagetarot`	vintage tarot style
`krea2_rainywindow`	rainy window style		

6 Your hardware WHY IT IS SLOW, AND WHAT THE ERRORS MEAN

On a 16 GB card this one fits. The artist is 12.2 GB, so it can sit on the card whole.

VRAM THE WORKBENCH

The graphics card's own memory. Whatever is sitting on it gets worked on at full speed, so this is the number that decides what is comfortable. This guide assumes 16 GB.

RAM THE SHELF

The PC's regular memory. Model files wait here when it is not their turn, and anything that does not fit on the workbench lives here. This guide assumes 32 GB.

Offloading THE SHUFFLE

When a model is bigger than the workbench, ComfyUI keeps the overflow on the shelf and swaps pieces on and off the card as each is needed. It is automatic. The result comes out the same. It only costs time.

The numbers

The default set is about **17 GB of model** in total: artist 12.2, reader 4.9, unpacker 0.2. They do not all sit on the card at once. They **take turns**: the reader goes first and steps aside, the artist does its 8 steps, the unpacker finishes. On 16 GB nothing has to spill at 1K. On 12 GB the artist is right at the edge, so a little of it spills and it runs slower. On 8 GB most of it spills. It still works, just slowly, and that is the case the 7.2 GB `nvfp4` file was made for if your card is an RTX 50.

Too slow, or out of memory? Pull these in order

1. **Lower the megapixels.** Back to 1. The biggest saving for one click.
2. **Turn Refine Prompt off.** The rewrite is extra waiting before the first step.
3. **Check the steps.** Turbo should say 8. If it says 20 or 50, that is your time.
4. **Smaller files.** On an RTX 50 card, the `nvfp4` Turbo file.
5. **Free the machine.** Close everything else that uses RAM or the graphics card.

What an error is telling you

`Red boxes when a workflow opens`
Missing custom nodes, or the workflow needs a newer ComfyUI. Update first, then let ComfyUI Manager install what is missing.

`"Value not in list" on a loader`
That file is not in the right models folder, or it has a different name than the workflow expects. Pick yours from the dropdown.

`"CUDA out of memory" / "Allocation on device"`
The card ran out of VRAM. Work down the list on the left.

`Dies after the last step finished`
Usually the VAE running out while unpacking. Lower the resolution.

`The whole PC crawls or freezes`
RAM is full and Windows is using the disk as memory. Close other programs, or move to smaller files.

The other words in this corner

CUDA / PyTorch THE PLUMBING

PyTorch is the math engine ComfyUI is built on. CUDA is NVIDIA's layer that lets PyTorch use the graphics card. Both come bundled with ComfyUI and you never touch them. You only meet them in messages: "CUDA out of memory" means the card is full, and a tag like `cu130` in the startup log is just the CUDA version PyTorch was built for.

Custom node AN ADD-ON

Every box on the canvas is a node. Custom nodes are boxes written by other people, installed through ComfyUI Manager. **Krea 2 itself needs none**: its nodes are built into ComfyUI. You only need a pack when a workflow you downloaded uses one. Each pack is code running on your PC, so stick to well-known ones.

rgthree A POPULAR PACK

A set of convenience nodes that shows up in a lot of shared workflows. The two you will meet most: a LoRA loader that stacks several LoRAs in one box, and switches that turn whole groups of a workflow on and off.

7 First run A KNOWN-GOOD STARTING POINT

Get one plain picture working with the official setup before you touch anything clever.

The recipe

1. **Update ComfyUI.** Older builds do not know the model.
2. **Open the built-in template.** Workflow menu, Browse Templates, search for Krea 2, pick the Turbo text to image one.
3. **Get the files.** ComfyUI lists what is missing and where it goes. It is the first three from section 4.
4. **Leave the defaults.** 1 megapixel, `8` steps, CFG `1`, euler, simple.
5. **Run it.** The first run is the slowest, because every file has to come off the disk.
6. **Run it again with Refine Prompt off.** Same seed. Now you know what the rewriter was adding.

The prompt

With Refine Prompt on, your prompt is a brief. Say the things you care about and let the rewriter fill in the rest. Anything you leave out, it will decide for you.

With it off, you are the rewriter. Long and specific beats short and vague: subject, setting, light, lens, mood, style.

Words in the picture: put the exact text in quotes, such as a book titled "Field Notes".

Style LoRAs: pick the style in the template and it adds the trigger words. Anywhere else, type them into the prompt yourself.